Homework 5

Information

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Goals

Homework:

Plot the time waveform of **all** the sounds used in your experiments (only 100ms of the signal), and their time-frequency decomposition (a.k.a. a spectrogram. Use MATLAB's spectrogram() function). (1 pts)

Load the EEG signal in MATLAB and use the trigger you generated in your s-function to separate the EEG response only when the sound was playing. Since the tone you played was modulated at a particular frequency, the best way to analyze it is in the frequency domain.
Take the Fourier transform of each channel, and start your analysis by averaging the magnitude of the FFT over all channels.
Plot the average magnitude. You can expect to see a peak at 40Hz in the EEG data. (1 pt)

If the length of the FFT is L, given the sampling frequency of your EEG signal (256Hz), what sample in FFT corresponds to the modulator frequencies you used (40Hz, 30 Hz)? (1 pt)

Use the following command to plot the one-sided spectrum: plot(linspace(0, FS/2,N),f(1:end/2)), and mark the modulator frequencies on the plots for each condition. Measure the normalized ASSR component by dividing the magnitude of the Fourier transform by its sum. Each Fourier coefficient now represents the *relative* power at that frequency, instead of *overall* power. (1 pts)

Next, we will determine how much data is needed to get a good ASSR measurement. To answer this important question, repeat your analysis for cumulative 30s intervals (i.e. first 30s, first 60s, first 90s, …). Display the relative magnitude of the ASSR as a function of signal duration.

At what duration can you reliably detect the presence of the ASSR? Does the signal get stronger the more data you include, or does it reach a plateau after a certain duration? Do you think you collected enough data? If not, how much more do you think you need to reach a plateau? (5 pts)

Measure the ASSR component separately for all channels at the modulator frequency (40Hz in condition 1, 30Hz in condition 2). Use the scalp map to show the relative magnitude of the ASSRs at all electrodes under each experimental condition (30Hz and 40Hz).
Which electrodes recorded the largest ASSR component? (1 pt)

Figure 1. screenshot of requirements

Task 1: Plot the sounds waveform in time and frequency domain

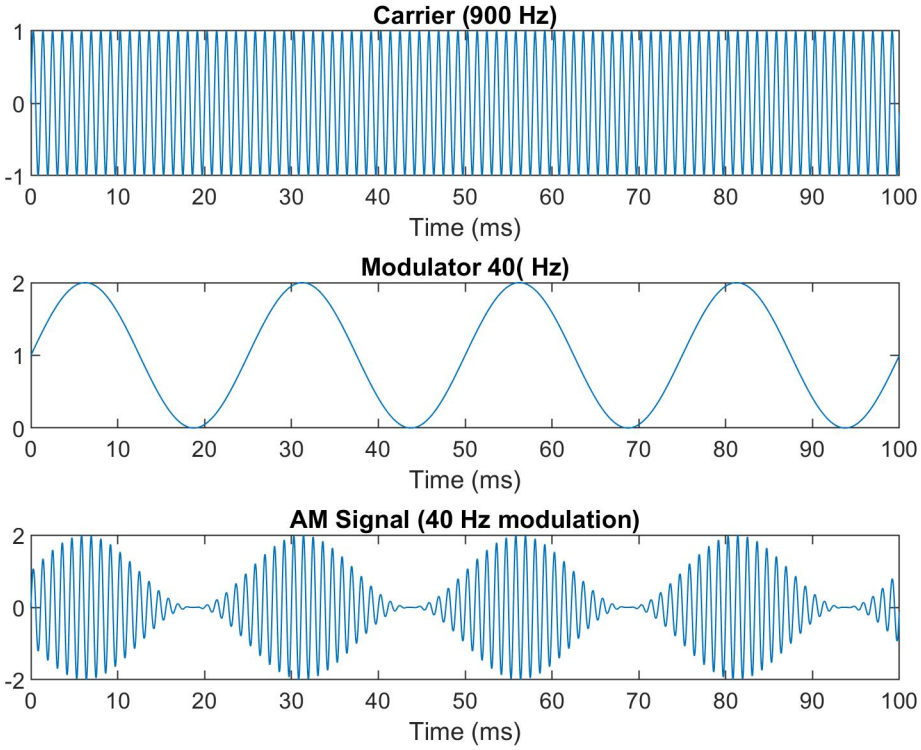


Figure 2. time domain (message frequency 40Hz)

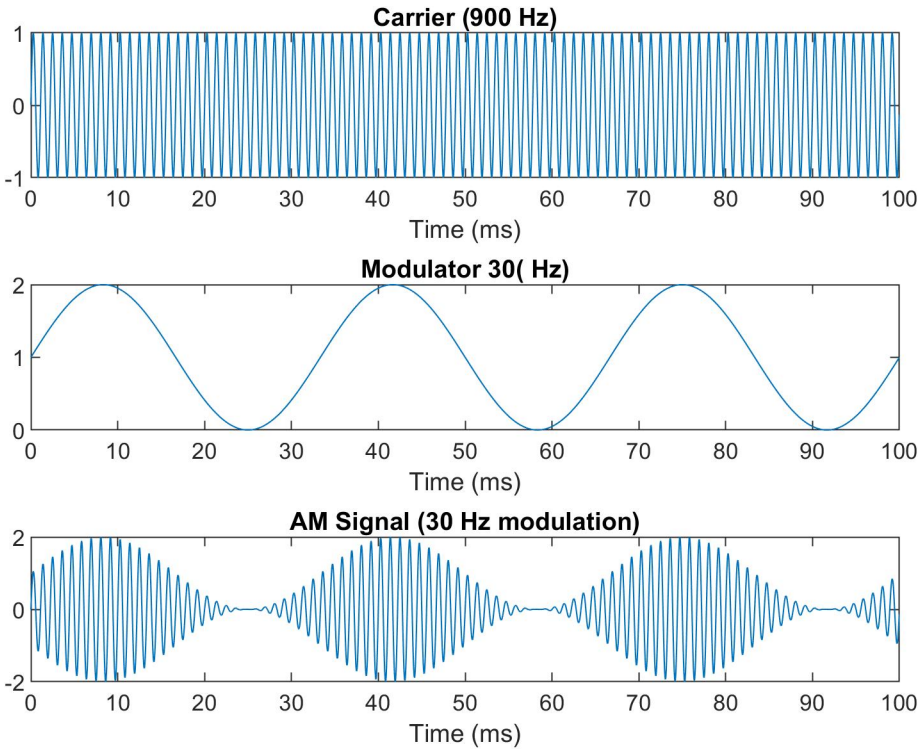


Figure 3. time domain (message frequency 30Hz)

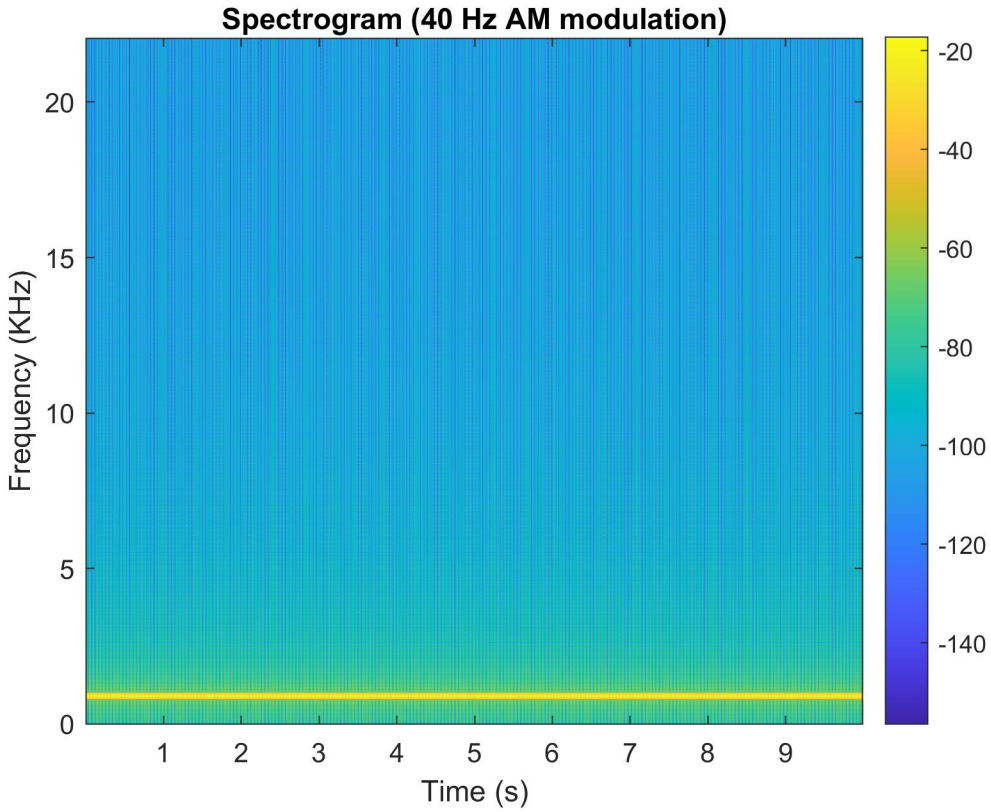


Figure 4. frequency domain (message frequency 40Hz)

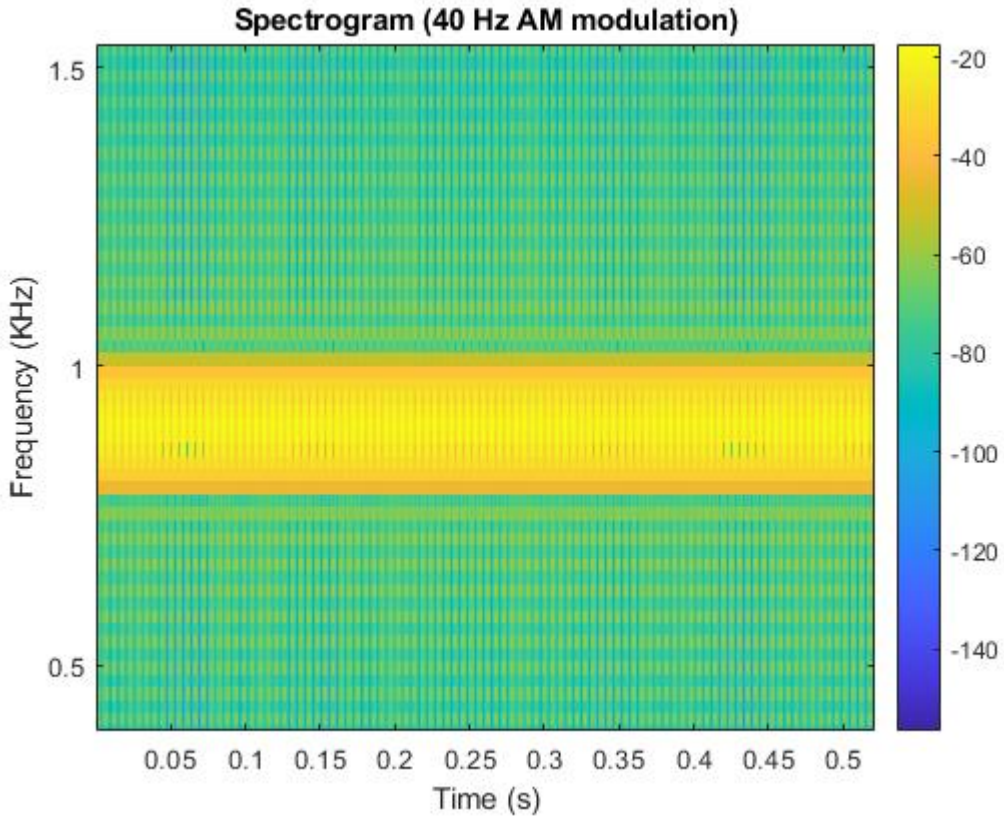


Figure 5. scaled frequency domain (message frequency 40Hz)

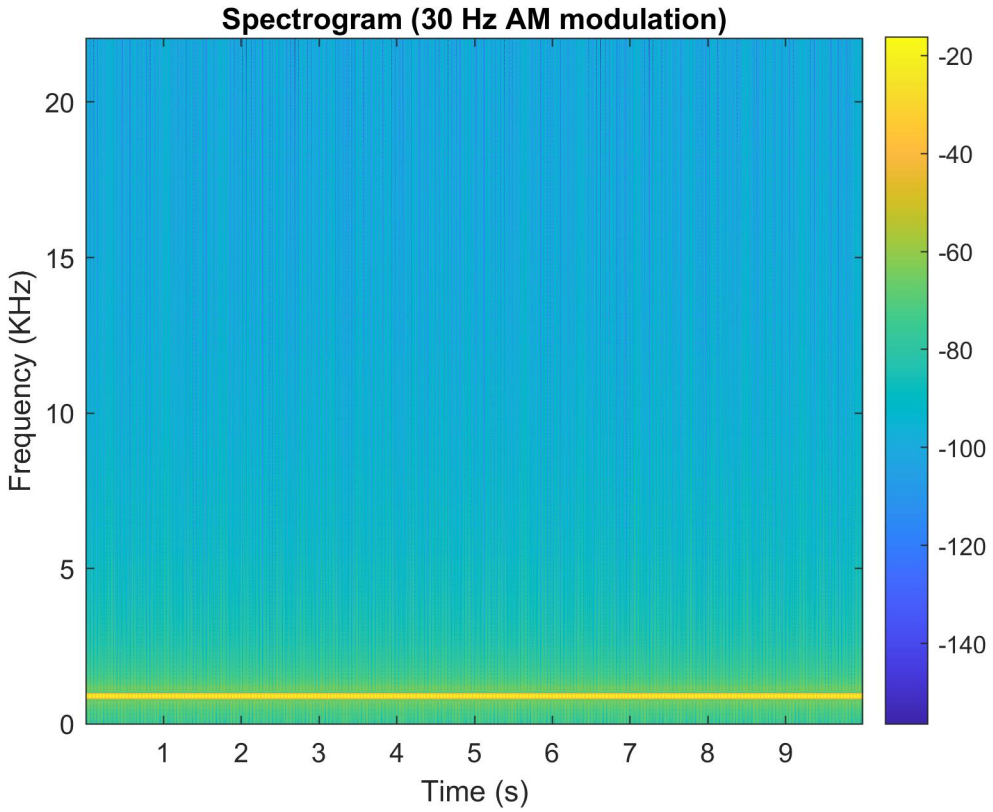


Figure 6. frequency domain (message frequency 30Hz)

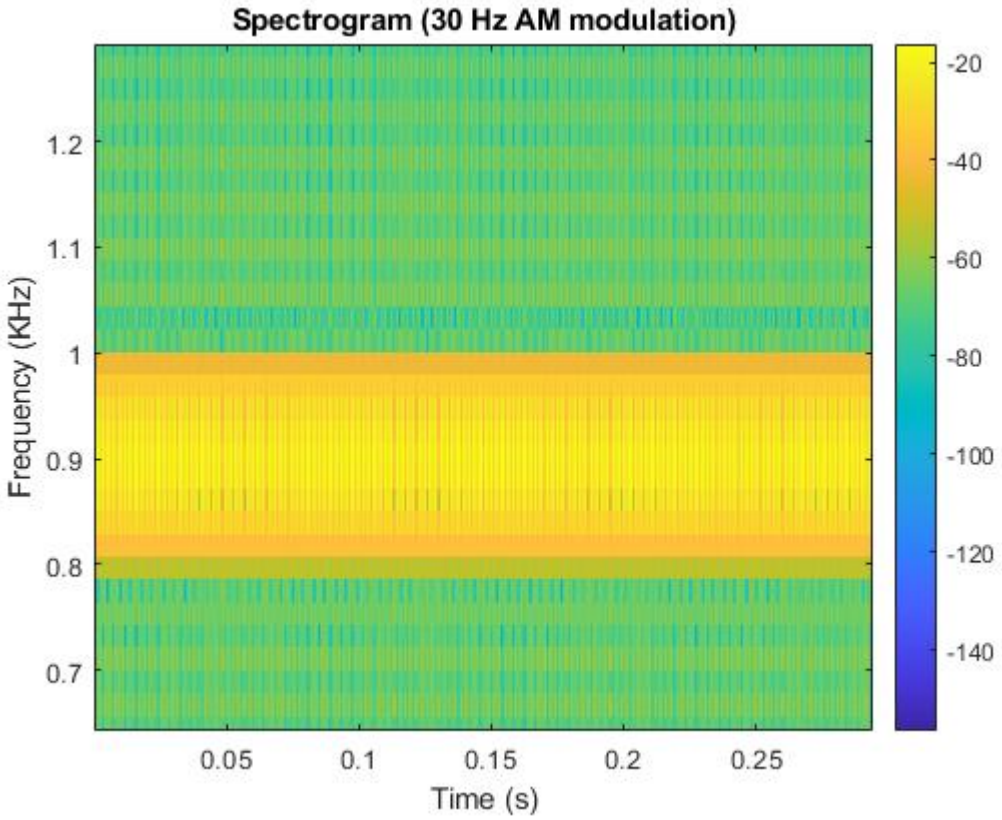


Figure 7. scaled frequency domain (message frequency 30Hz)

Task 2: Plot the average magnitude

Figure 8 shows the FFT results: the first and third subplots exhibit the average amplitude-frequency responses across all 7 electrodes and the second and fourth subplots exhibit the phase-frequency responses. It's evident that there is a clear 40Hz component in data1 and 30Hz component in data2.

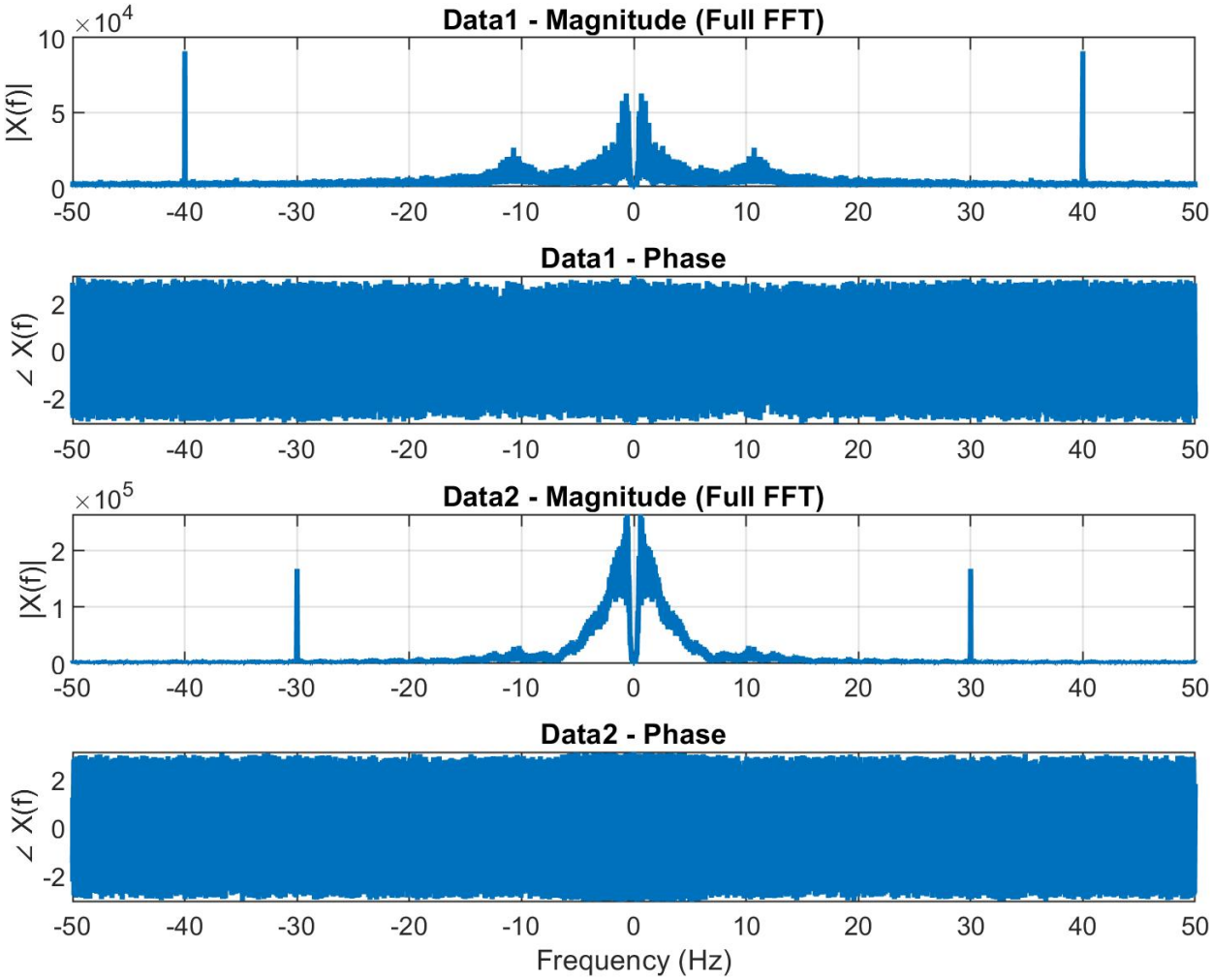


Figure 8. average magnitude

Task 3: Calculation the sample

For an FFT of length ***L*** and sampling frequency f_s , the sample index ***k*** corresponding to a frequency ***f*** is:

$$k = \frac{f \cdot L}{f_s}$$

Thus:

$$\begin{cases} k_{40} = \dfrac{5L}{32} \\ k_{30} = \dfrac{15L}{128} \end{cases}$$

Task 4: Plot the one-sided spectrum

Figure 9 shows the one-sided spectrum of both trials: the blue line shows the average one-sided amplitude-frequency response when the message frequency is 40Hz and the orange line shows the result of 30Hz message frequency.

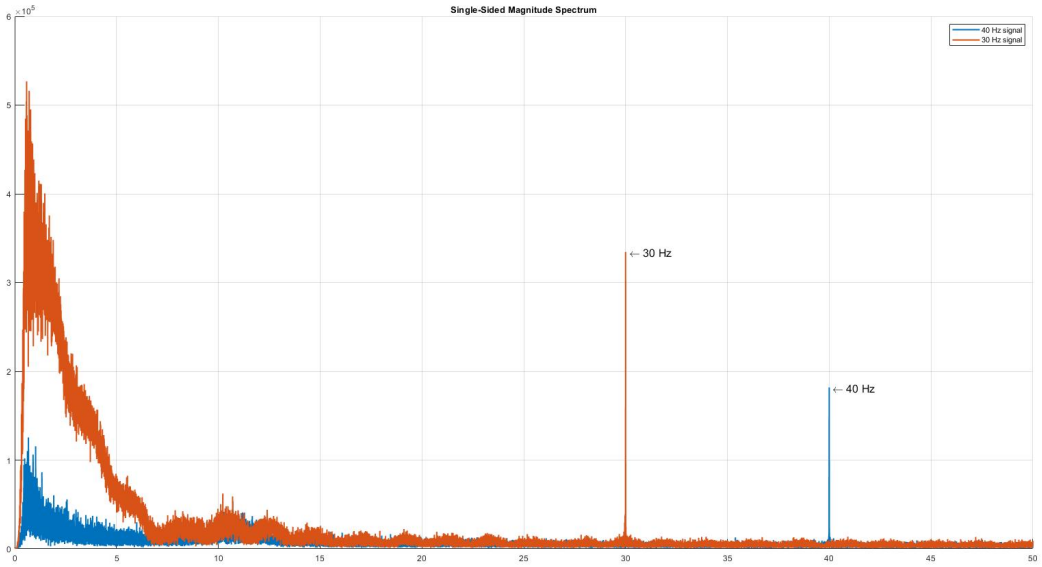


Figure 9. one-sided spectrum

Task 5: Display the relative magnitude

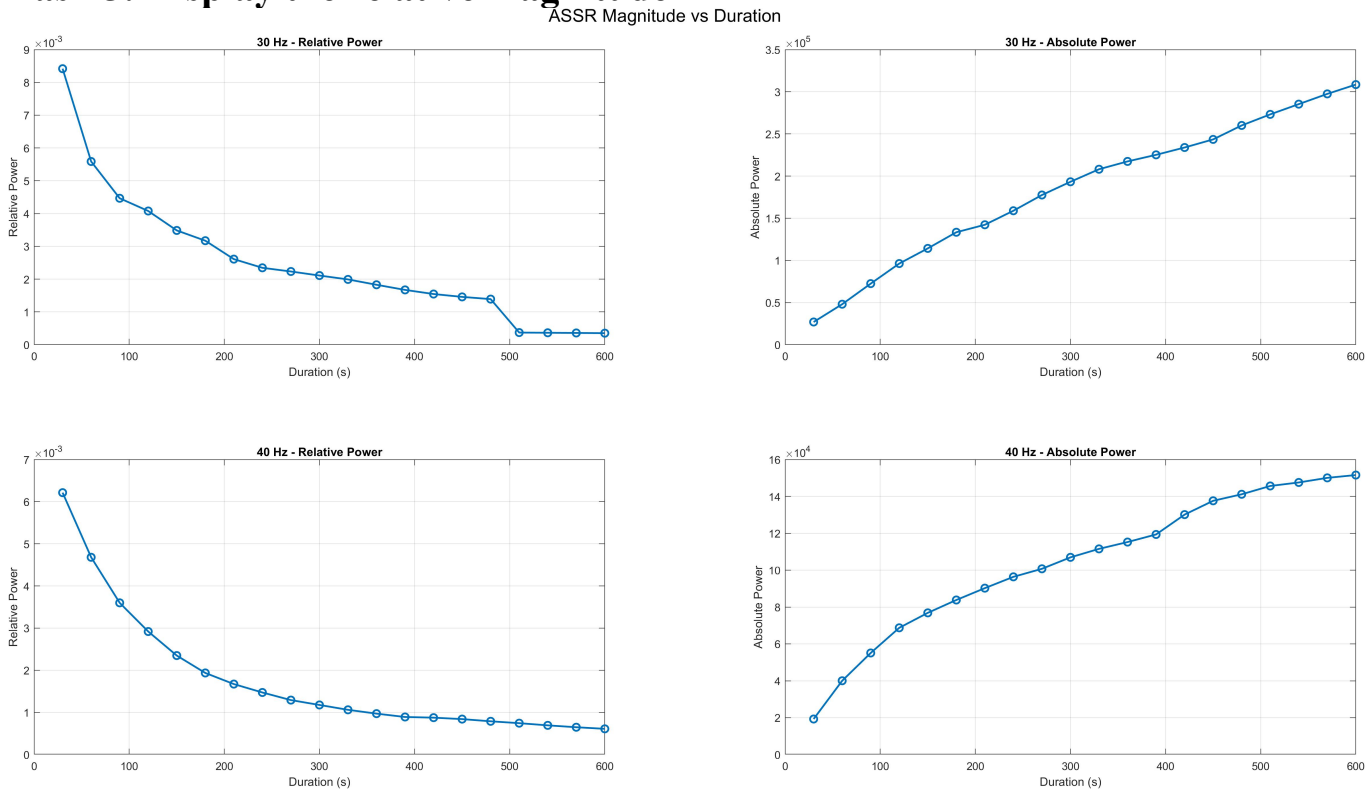


Figure 10. ASSR magnitude vs duration

Figure 10 shows the relative power (left) and absolute power (right). Although the absolute powers of 30 Hz and 40 Hz increase as the time interval grows, the relative powers decrease. A possible reason is that a longer time-domain sequence causes a higher-resolution frequency spectrum. As the resolution improves, more frequency components surrounding the target frequency are separated out, so the energy of the target frequency becomes less dominant relative to the whole spectrum.

To tackle with the impact caused by resolution, Welch's method is used to compute the power spectrum, from which we extract the power at the target modulation frequency. The surrounding frequency samples serve as the noise floor, and the SNR is calculated in dB. Finally, the SNR values are plotted in Figure 11.

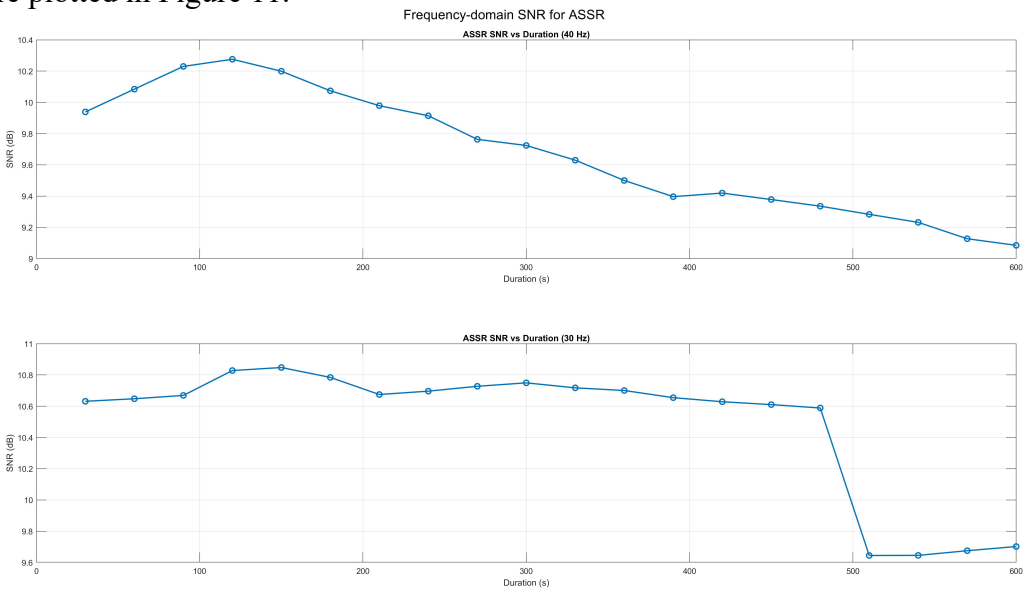


Figure 11. ASSR SNR compared to surrounding samples

According to the results, the SNR of data1 (40Hz) reaches its peak at 120s and gradually drop from 120s to 600s. However the the SNR of data2 (30Hz) peaks at150s and stays at a plateau before a sudden plunge at 510s. Based on this behavior, it appears that sufficient data have already been collected for reliable ASSR detection.

Task 6: Scalp map

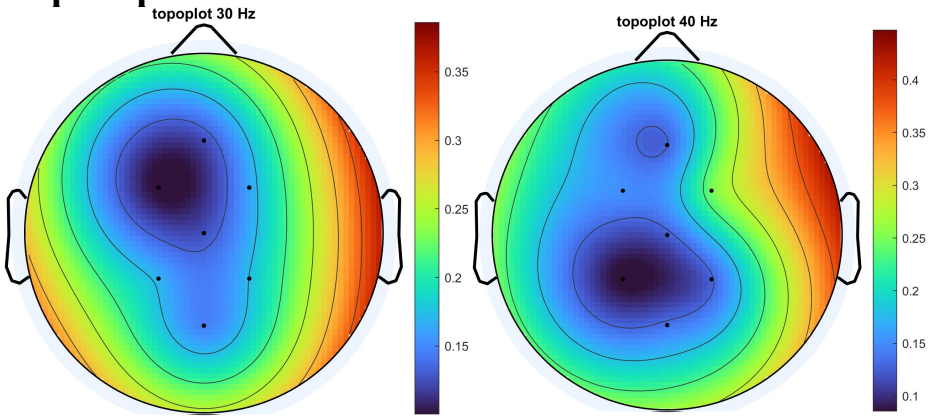


Figure 12. scalp map (left: 40Hz, right: 30Hz)

Figure 12 use the scalp map to show the relative magnitude of the ASSRs at all electrodes under each experimental condition(left: 40Hz, right: 30Hz). According to the subplots, CP2 and FC2 (right brain) recorded the largest ASSR component.